



ALLEN INSTITUTE *for*
BRAIN SCIENCE

MapMyCells

Describe Your Neurons workshop

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What is MapMyCells?

- Web tool
 - <https://brain-map.org/bkp/analyze/mapmycells>
- Open source Python library
 - https://github.com/AllenInstitute/cell_type_mapper
- Takes anything that “looks like” cell-by-gene data and labels each cell according to supported BICAN cell type taxonomies
 - Whole Mouse Brain (Yao et al. 2023)
 - Whole Human Brain (Siletti et al. 2023)
 - Cross-species Basal Ganglia (Johansen et al. 2025)
 - SEA-AD medial temporal gyrus (Gabbito et al. 2024)

How do I use MapMyCells?

- Accepts .h5ad files and CSV files.
 - .h5ad format defined by anndata library:
<https://anndata.readthedocs.io/>
- Ideally, rows are cells and columns are genes.
 - The web tool will do its best to transpose your data if rows are genes.
- For web tools, files must be less than 2 GB in size (can be gzipped CSVs).

That's it!

Code automatically uses tables published by NCBI to align genes with reference taxonomy.

Demonstration

Rhesus macaque data from CellXGene

<https://cellxgene.cziscience.com/collections/e1fa9900-3fc9-4b57-9dce-c95724c88716>

MapMyCells algorithm (in brief)



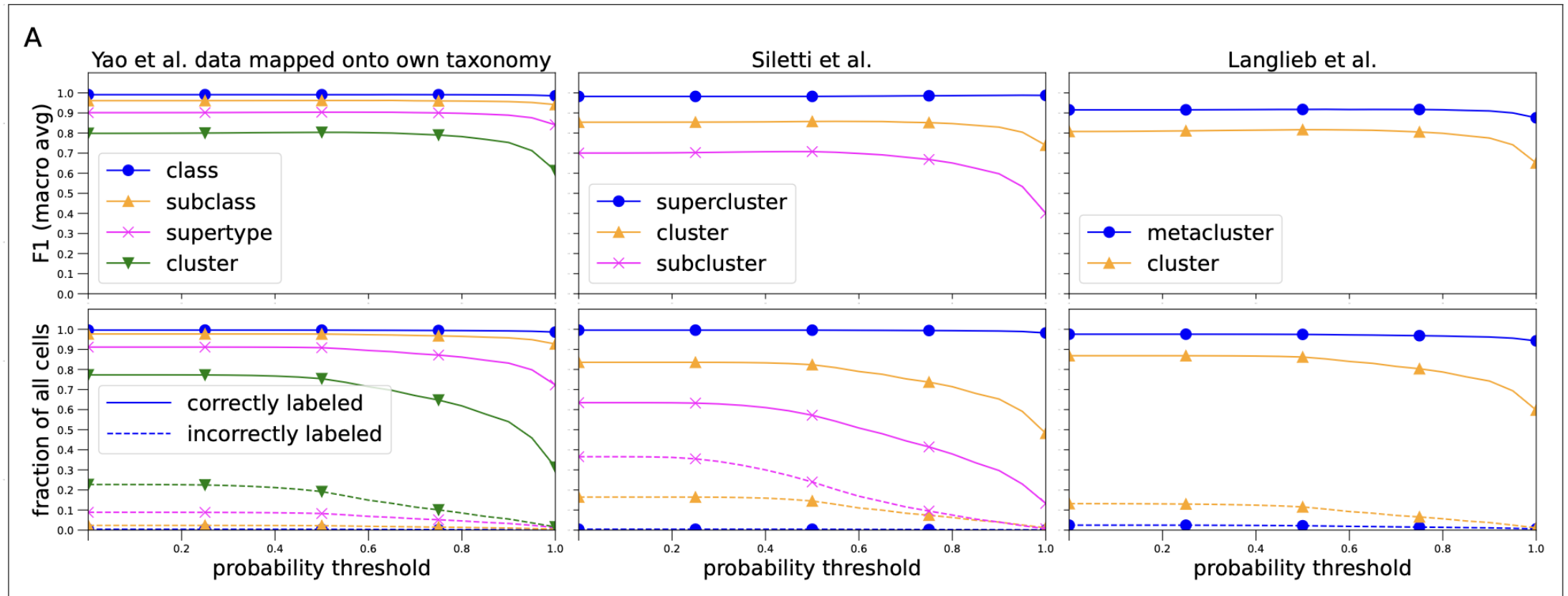
link to webinar with more
detailed description

- Correlate each cell against cell types using a pre-computed set of marker genes.
- Walk down the hierarchy of the taxonomy, first choosing a class, then a subclass, then a supertype.... etc.
- Perform each mapping many times, using a random subset of marker genes each time.
- Select the cell type that was chosen by the plurality of random iterations.

Performance

MapMyCells pre-print

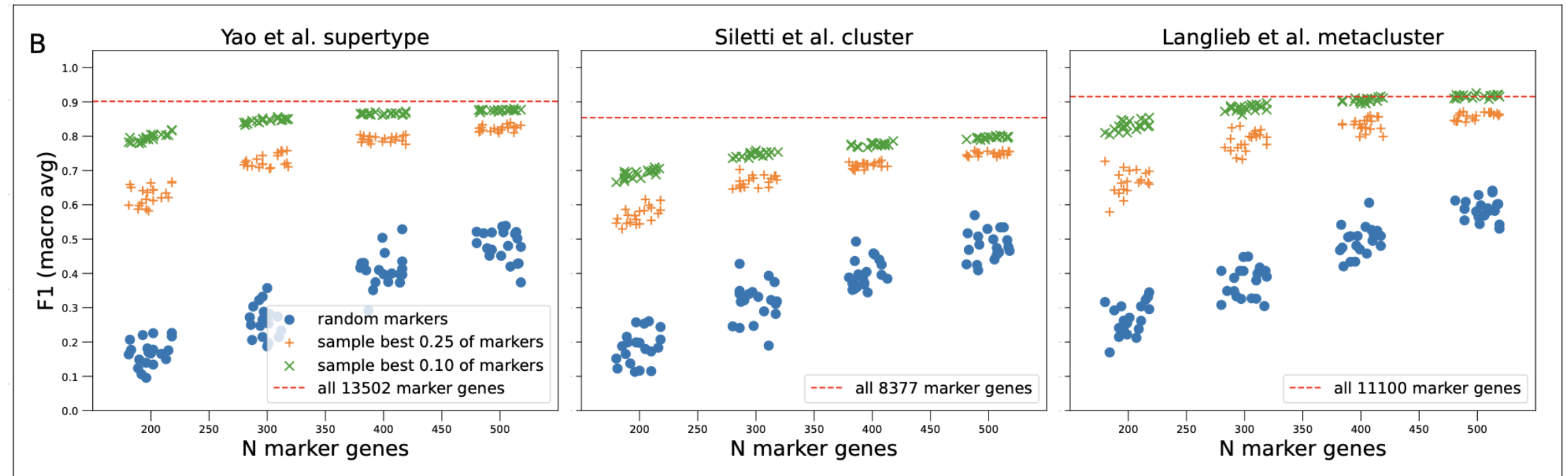
<https://doi.org/10.64898/2026.03.06.710160>



Performance

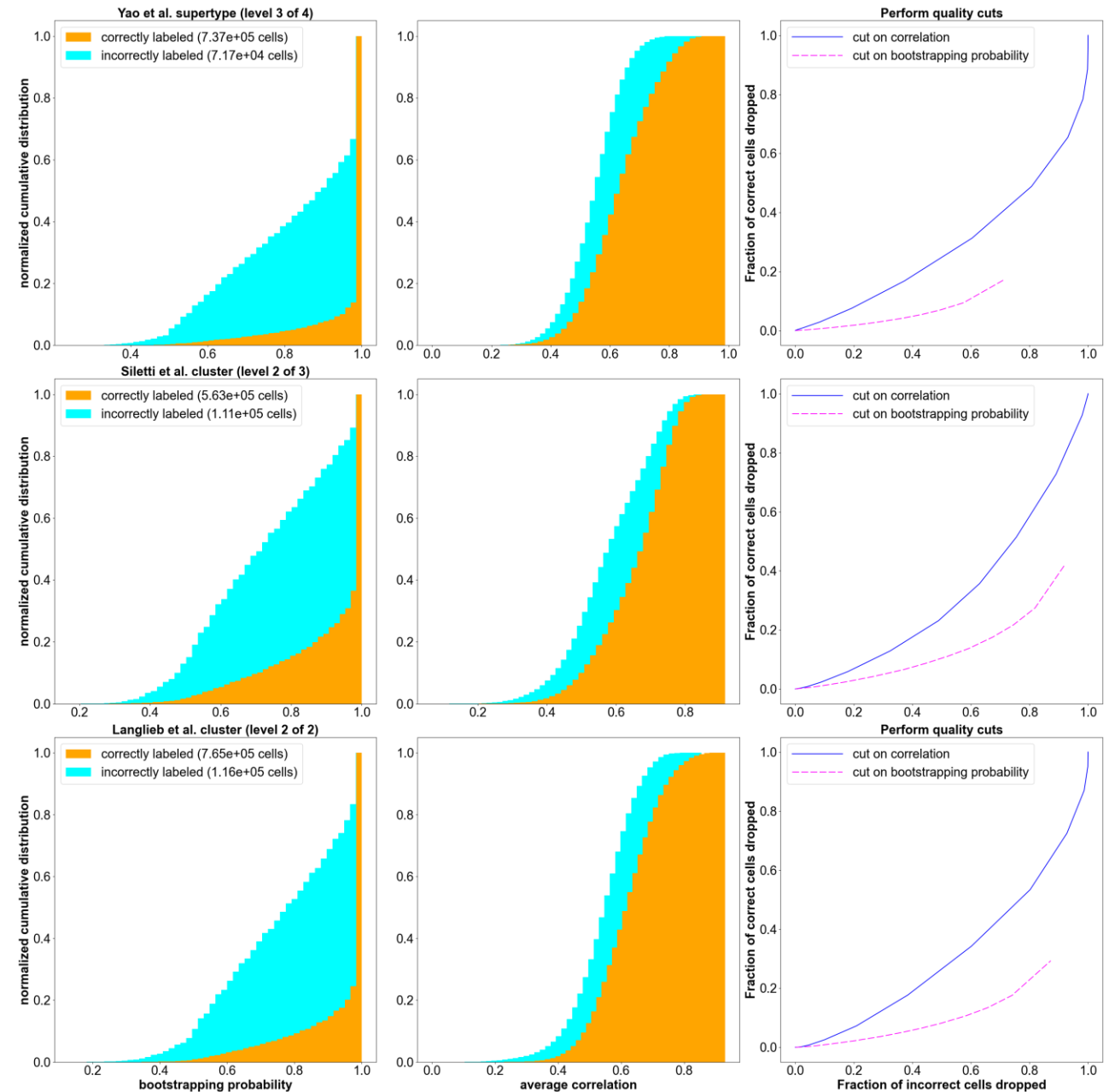
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Performance metrics

- **average correlation:** the correlation coefficient between cell and cell type averaged over iterations that chose that cell type.
- **bootstrapping probability:** fraction of iterations that chose cell type.



Visualize results of demo

https://github.com/AllenInstitute/SPOCTT/blob/main/2026_Describe_Your_Neurons_Workshop/MapMyCells_demo.ipynb

